



CHARAN METTUR SELVARAJ

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Master's Student – Information & Communication Systems

Open to Relocation | Available Immediately

PROFESSIONAL SUMMARY

Master's student in Information and Communication Systems at Technische Universität Chemnitz with professional experience in telematics, connected vehicle systems, and GNSS-based fleet tracking solutions. Skilled in automotive diagnostics, vehicle communication systems, REST APIs, and real-time telemetry workflows with hands-on experience in Python, C++, FastAPI, and Streamlit. Worked on projects involving ADAS prototypes, computer vision, AI-based driver monitoring, EV battery health prediction, and GenAI-powered vehicle diagnostic assistants using RAG, ChromaDB, Ollama LLM, and semantic retrieval pipelines. Strong interest in automotive connectivity, intelligent mobility systems, vehicle diagnostics, ADAS, AI-driven automotive applications, and telecom-integrated connected vehicle technologies.

SKILLS

- **Programming & Software Development:** Python, C/C++, MATLAB, SQL, React.js, Object-Oriented Programming (OOP)
- **Backend & Web Technologies:** FastAPI, REST APIs, JSON, HTTP Methods, Streamlit, Dashboard Development
- **Automotive & ADAS:** Vehicle Diagnostics, OBD-II Protocols, ECU Communication, CAN Basics, Automotive Telematics, GNSS-Based Vehicle Tracking, Geofencing, ADAS Concepts, Driver Monitoring Systems, Lane Detection, ISO 26262 Fundamentals, AUTOSAR Basics
- **GNSS & Sensor Fusion:** GPS, GLONASS, Galileo, Vehicle Positioning, Kalman Filter (KF), Extended Kalman Filter (EKF), Geospatial Processing
- **Telecom & Connectivity:** 3GPP Basics, NAS Procedures, AT Commands, Cellular Modem Communication, TCP/IP, Ethernet/IP, V2X Communication Fundamentals
- **AI, Machine Learning & GenAI:** PyTorch, YOLOv8, OpenCV, MediaPipe, Scikit-learn, Random Forest, LLM Applications, Retrieval-Augmented Generation (RAG), Ollama, ChromaDB, Semantic Search
- **Tools & Platforms:** Linux, Git, Jira, MongoDB (Basic), CST Studio Suite, Android Debug Bridge (ADB)

EDUCATION

Master of Science - Information and Communication Systems- October 2024 – Present

Technische Universität Chemnitz, Germany

- Relevant Coursework: V2X communication, Multisensorial Systems, Computer Vision, Digital Signal Processing, Advanced Mobile Networks

Bachelor of Engineering – Information Technology - 2019 – 2023

Sri Krishna College of Technology, Coimbatore, India

- Focus areas: Embedded Systems, IoT, Computer Networks, Wireless Communication

PROFESSIONAL EXPERIENCE

Telematics Systems Engineer (Core Functional Development) - ZED Digital, India | 2022 - 2024

- Contributed to IoT-based truck tracking and fleet monitoring solutions using GNSS and OBD-II data, supporting real-time tracking for 100+ vehicles.
- Developed core logic for vehicle positioning and AVL systems, improving location update efficiency by ~25%.
- Implemented geofencing functionality, enabling real-time alert generation with <2 sec latency for zone entry/exit events.
- Integrated REST APIs (HTTP/JSON) to process live telemetry data, reducing data retrieval latency by ~20%.
- Supported backend workflows for GNSS data ingestion and validation, improving data reliability and consistency by ~30%.
- Assisted in React.js dashboard development, enhancing fleet visibility and reducing manual monitoring effort by ~40%.

PROJECTS

Waveguide-Based RF Measurement & Material Characterization

Vector Network Analyzer (VNA), RF Measurement, MATLAB

- Investigated electromagnetic wave propagation in a rectangular waveguide using different materials under test across multiple frequency ranges (GHz band)
- Measured transmission, reflection, and S-parameters using a Vector Network Analyzer (VNA) with high measurement precision (± 0.5 dB)
- Evaluated the influence of PMMA, PET, aluminum, and air-gap discontinuities on RF signal propagation, comparing 4+ material configurations
- Analyzed frequency-domain behavior and interpreted measurement data to compare material characteristics and shielding performance, improving analysis accuracy by ~25% through MATLAB-based processing

Automated NAS Procedures Using AT Commands

Python

- Automated basic NAS procedures such as network registration and call control using AT commands, reducing manual testing effort by ~50%
- Developed Python scripts to execute AT commands over serial communication based on 3GPP TS 27.007, handling 50+ command executions per test cycle
- Successfully monitored registration status, call states, signal quality, and modem responses with ~40% faster validation cycles
- Gained practical understanding of cellular modem operation, NAS procedures, and telecom protocols through real-time testing and automation workflows

Sensor Fusion for GNSS-Based Vehicle Positioning using Kalman and Extended Kalman Filters

Python

- Improved vehicle positioning accuracy using Kalman Filter (KF) and Extended Kalman Filter (EKF), achieving ~30–40% reduction in positioning error (ADE).
- Processed real GNSS vehicle data (10,000+ data points) and implemented KF/EKF models in Python.
- Reduced positioning noise and improved tracking continuity; EKF showed better accuracy through ADE evaluation with ~35% noise reduction and ~25% improvement in trajectory stability.
- Learned sensor fusion, GNSS data processing, and vehicle tracking concepts relevant to telematics through hands-on model evaluation and performance analysis.

Real-Time Vehicle Tracking with Circular Geofencing

Python, FastAPI, Streamlit, Leaflet (Folium), HTTPS, GPS

- Built a real-time vehicle tracking and geofencing system for fleet monitoring supporting 10+ concurrent vehicle simulations.
- Used FastAPI, Streamlit, GPS data, and Haversine-based circular geofencing with <2 seconds alert response latency.
- Generated live vehicle tracking and entry/exit alerts on an interactive dashboard handling 100+ location updates per minute.
- Gained hands-on experience with REST APIs, telemetry workflows, geospatial processing, and dashboard visualization, improving data visualization clarity by ~20%.

Ceramic Chip Antenna Design – 2.4 GHz ISM Band

CST Studio Suite, MATLAB

- Designed a compact ceramic chip antenna operating in the 2.4 GHz ISM band, relevant to vehicle telematics communication modules (Bluetooth / short-range wireless links), achieving resonance at ~2.4 GHz with stable frequency response
- Performed antenna modeling, impedance matching ($50\ \Omega$), and S-parameter analysis to ensure reliable RF performance with return loss < -10 dB
- Evaluated return loss, bandwidth, radiation pattern, and realized gain to assess suitability for embedded automotive devices with ~100–150 MHz bandwidth coverage
- Validated antenna behavior through electromagnetic simulations, focusing on performance consistency in compact system designs with <5% variation across simulations

USIM File System Emulator

Python

- Implemented a protocol-driven USIM file system emulator to understand SIM-based authentication and data access in connected systems, supporting multiple SIM file operations (10+ file types)
- Designed structured request–response handling and file access logic following telecom protocol behavior, improving processing efficiency by ~30%
- Simulated SIM file operations relevant to cellular-enabled devices used in telematics and IoT systems, enabling realistic protocol behavior testing scenarios
- Strengthened understanding of mobile network integration, device identity and communication workflows through end-to-end simulation and testing

Vehicle Diagnostics Tool (OBD-II Simulation)

C++ | TCP/IP | JSON | OBD-II Protocols

- Created a simulation-based tool to emulate vehicle diagnostics and ECU communication supporting 20+ OBD-II parameters (speed, RPM, fault codes).
- Developed a modular C++ application using TCP/IP, JSON, and OBD-II PIDs, achieving <100 ms response latency.
- Simulated speed, RPM, fault conditions, and diagnostic responses in real time, handling multiple concurrent diagnostic signals.
- Gained practical knowledge of OBD-II protocols, ECU communication, and automotive software design, improving debugging efficiency by ~30%.

Gen AI-Based Vehicle Diagnostic Assistant (RAG System)

Python, FastAPI, Streamlit, ChromaDB, Ollama (LLM), REST APIs

- Built an end-to-end GenAI diagnostic system to analyze vehicle DTCs, symptoms, and sensor inputs, leveraging a knowledge base of 1000+ automotive fault records.
- Implemented a full RAG pipeline using ChromaDB embeddings and semantic search to retrieve relevant automotive fault knowledge, improving diagnostic relevance by ~35%.
- Developed a scalable FastAPI backend handling input preprocessing, retrieval, and LLM-based reasoning with structured diagnostic outputs, achieving <2 seconds response time.
- Designed an interactive Streamlit dashboard for real-time user input and visualization of fault summaries, probable causes, severity, and repair actions, supporting real-time interaction workflows.
- Integrated Ollama LLM to generate context-aware diagnostic recommendations by combining retrieved knowledge with user-provided data, reducing manual troubleshooting effort by ~40%.

EV Battery Health Prediction Dashboard

Python, Machine Learning, FastAPI, Streamlit, Scikit-learn, Random Forest, REST APIs

- Developed an end-to-end Machine Learning system to predict EV battery State of Health (SOH) using battery usage, charging behavior, temperature, mileage, and electrical parameters, processing 1000+ battery data samples.
- Trained and optimized a Random Forest Regressor on structured battery datasets to estimate SOH %, classify battery condition, and identify degradation risk levels with ~85–90% prediction accuracy.
- Built a scalable FastAPI backend for real-time prediction APIs, input validation, preprocessing, and recommendation generation with <1 second API response time.
- Designed an interactive Streamlit dashboard for battery analytics, enabling users to input battery data and visualize SOH, risk score, and maintenance insights, supporting real-time user interaction workflows.
- Implemented rule-based maintenance recommendations to support predictive maintenance and improve battery lifespan in electric vehicles, reducing manual analysis effort by ~30%.

AI-Based Lane Detection & ADAS Warning System

Python, OpenCV, YOLOv8, PyTorch, Streamlit, Computer Vision

- Developed an AI-powered ADAS prototype for road-video analysis using Computer Vision and Deep Learning techniques, processing real-time video streams (~20–25 FPS).
- Implemented real-time lane detection using OpenCV with ROI masking, edge detection, Hough Transform, and lane smoothing, achieving ~90% lane detection accuracy under standard conditions.
- Integrated YOLOv8 (PyTorch-based) object detection to identify vehicles, pedestrians, traffic lights, and road obstacles with ~85%+ detection precision across multiple classes.

- Built ADAS warning logic to generate lane departure alerts, steering guidance, and surrounding object awareness with <1 second alert response time.
- Designed an interactive Streamlit dashboard for live video processing, AI visualization, and real-time driving safety metrics, enabling continuous monitoring and visualization.

AI-Based Driver Drowsiness Detection System

Python, OpenCV, MediaPipe, Streamlit, Computer Vision, NumPy

- Developed a real-time driver monitoring system using OpenCV and MediaPipe for facial landmark detection and fatigue analysis, processing live video streams (~20–25 FPS).
- Implemented eye closure detection using Eye Aspect Ratio (EAR) and blink tracking with NumPy-based computations, achieving ~92% detection accuracy.
- Integrated yawning detection (MAR) and head pose estimation to enhance robustness of fatigue detection, reducing false positives by ~25%.
- Built a drowsiness scoring, alert system (visual + audio), and Streamlit dashboard with event logging for driver safety monitoring, enabling <1 second real-time alert response.

CERTIFICATIONS

- PCEP – Certified Entry-Level Python Programmer
- Software Defined Vehicles – Digital.auto - Coursera
- Prompt Engineering with Vertex AI (Google Cloud)

LANGUAGES

- English – Fluent
- German – B1 (B2 in Progress)
- Tamil – Native